



BITS Pilani

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CE F231 Fluid Mechanics

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Lectures 31 and 32: Viscous Flow

Learning Outcome

- What is viscous flow?
- Classification of flow as viscous/laminar flow based on Reynold's number
- Shear stress distribution, velocity distribution, and head loss (Hagen-Poiseuille equation) for viscous flow in circular pipes
- Shear stress distribution, velocity distribution, head loss for viscous flow in between parallel plates

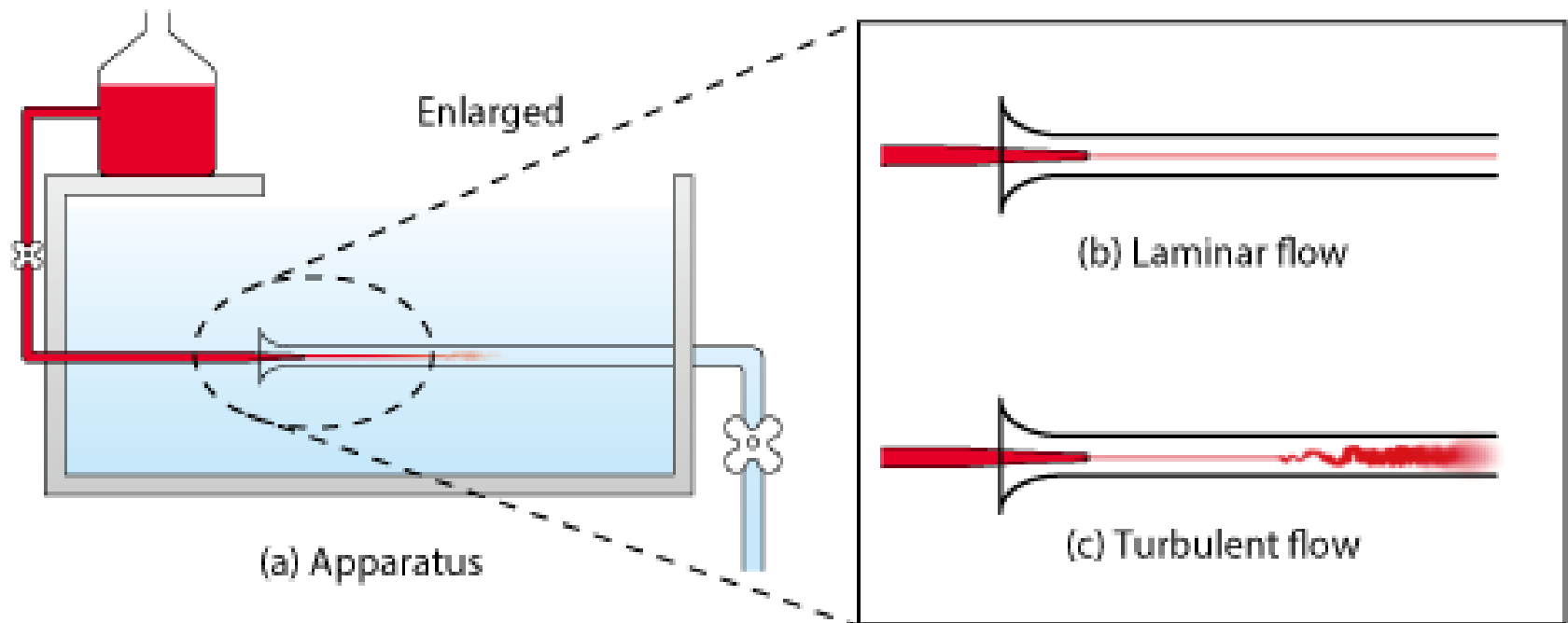
Viscous flow

- Two fluid layers move relative to each other → frictional force develops → depends on viscosity
- Viscosity → property due to which fluid opposes relative motion between different layers
- Viscous flow → fluid flow in which frictional/viscous effects are significant
- Viscous flow is observed in
 - Liquids with high viscosity
 - Low velocity

laminar flow



Reynold's experiment



$$Re = \frac{\text{Inertial Force}}{\text{Viscous Force}} = \frac{\rho V D}{\mu}$$

For pipe flow

Re < 2000: Laminar / Viscous

2000 < Re < 4000: Transitional

Re > 4000: Turbulent

Reynold's experiment



$$Re = \frac{\text{Inertial Force}}{\text{Viscous Force}} = \frac{\rho V D}{\mu}$$

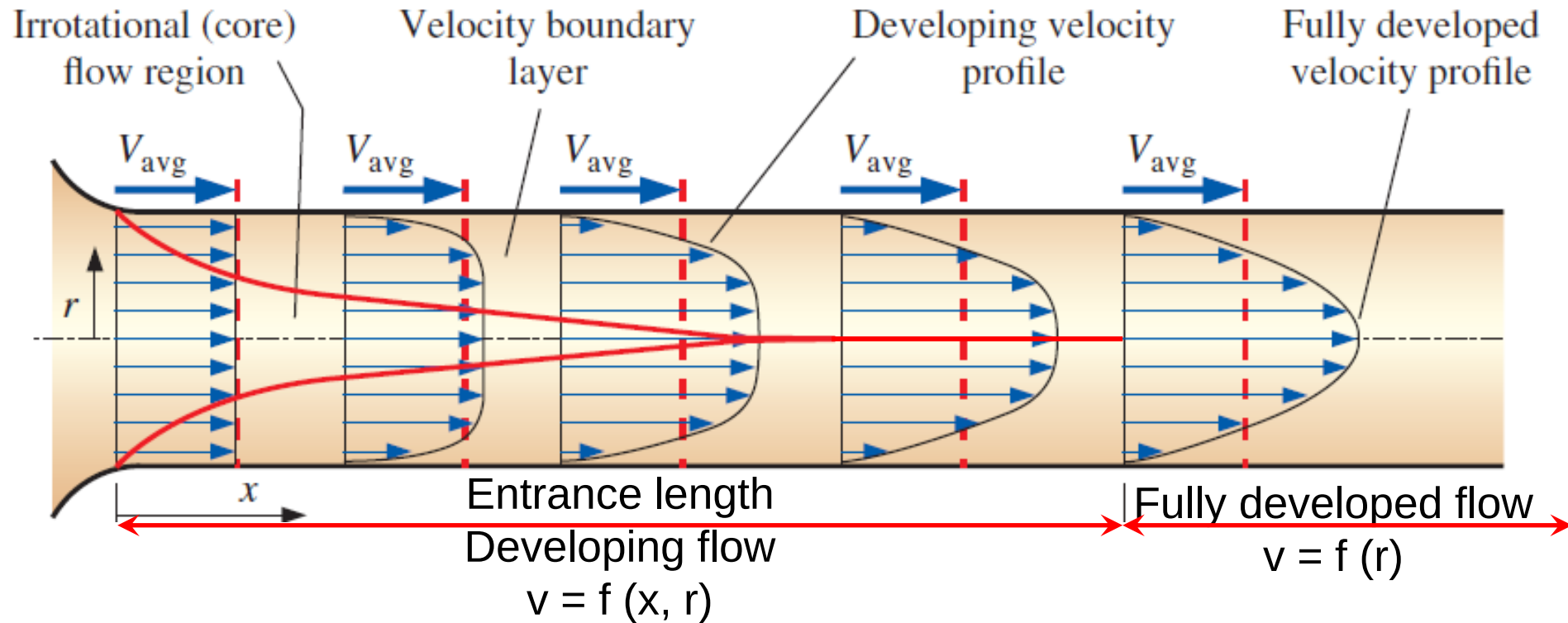
For pipe flow

Re < 2000: Laminar / Viscous

2000 < Re < 4000: Transitional

Re > 4000: Turbulent

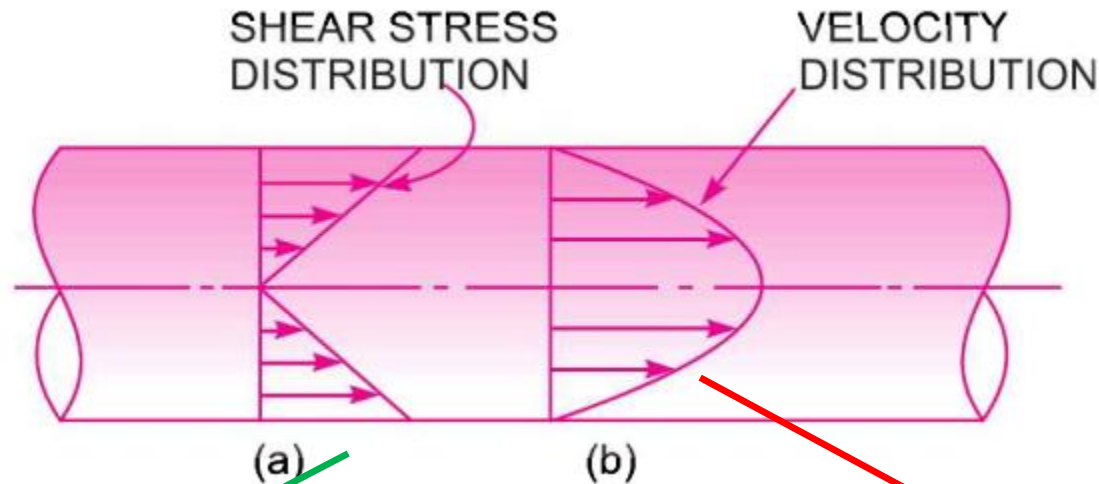
Boundary layer



- The boundary layer starts growing on the wall at the entrance of the pipe.
- Gradually it becomes thicker in the downstream direction.
- The flow becomes fully developed when the boundary layers from the wall meet at the axis of the pipe.

Viscous flow through circular pipes

Shear stress and velocity distribution profiles



$$\tau = -\frac{\partial p}{\partial x} \frac{r}{2}$$

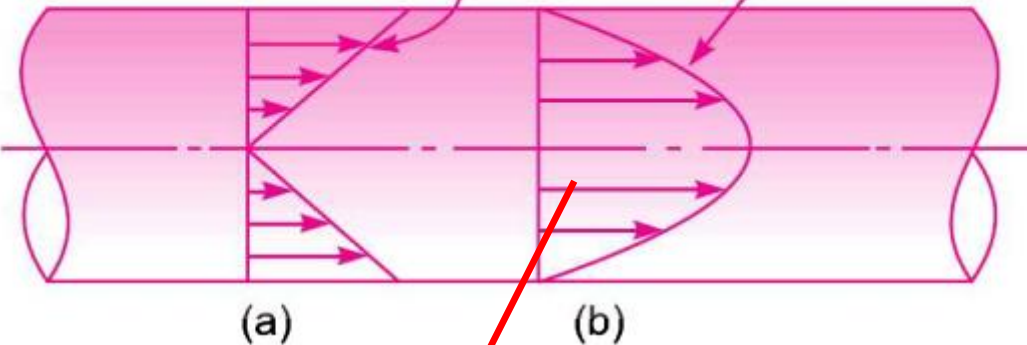
$$\begin{aligned} u &= \frac{1}{4\mu} \frac{\partial p}{\partial x} r^2 - \frac{1}{4\mu} \frac{\partial p}{\partial x} R^2 \\ &= -\frac{1}{4\mu} \frac{\partial p}{\partial x} [R^2 - r^2] \end{aligned}$$

Viscous flow through circular pipes

Relation between maximum velocity and mean velocity

SHEAR STRESS
DISTRIBUTION

VELOCITY
DISTRIBUTION

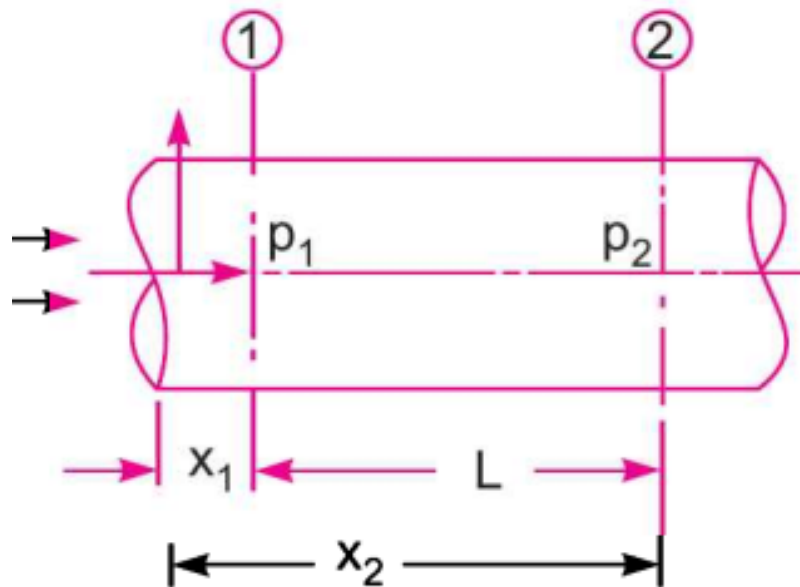


$$\frac{U_{\max}}{\bar{u}} = \frac{-\frac{1}{4\mu} \frac{\partial p}{\partial x} R^2}{\frac{1}{8\mu} \left(-\frac{\partial p}{\partial x}\right) R^2} = 2.0$$

$$\begin{aligned} u &= \frac{1}{4\mu} \frac{\partial p}{\partial x} r^2 - \frac{1}{4\mu} \frac{\partial p}{\partial x} R^2 \\ &= -\frac{1}{4\mu} \frac{\partial p}{\partial x} [R^2 - r^2] \end{aligned}$$

Viscous flow through circular pipes

Head loss estimation using Hagen-Poiseuille equation

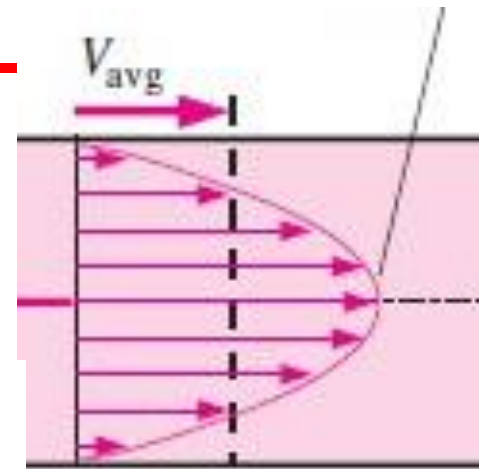
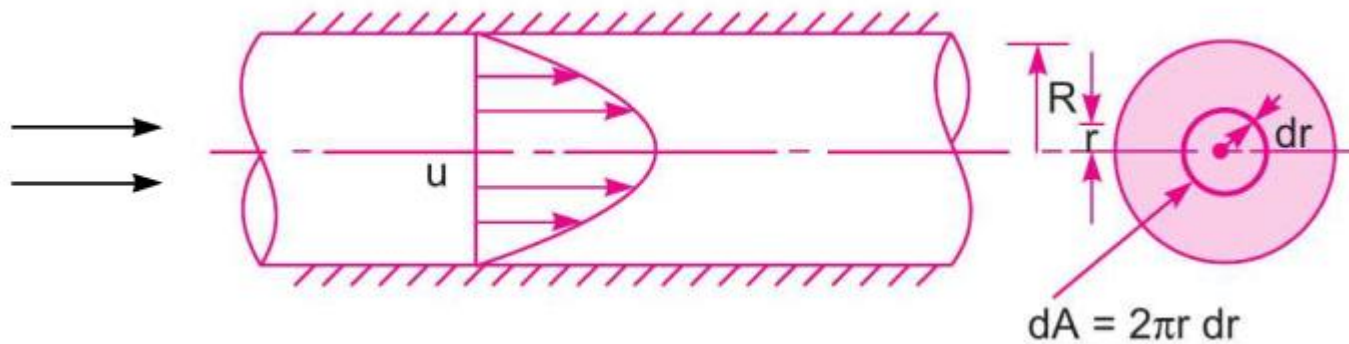


$$\frac{p_1 - p_2}{\rho g} = h_f = \frac{32\mu \bar{u} L}{\rho g D^2}$$

Viscous flow through circular pipes

Momentum correction factor

$$\beta = \frac{\text{Momentum per second based on actual velocity}}{\text{Momentum per second based on average velocity}}$$



$$u = \frac{1}{4\mu} \left(-\frac{\partial p}{\partial x} \right) (R^2 - r^2)$$

$$\begin{aligned} \text{Momentum of fluid through ring /sec} &= \text{mass} \times \text{velocity} \\ &= \rho \times dQ \times u = \rho \times 2\pi r dr \times u \times u = 2\pi\rho u^2 r dr \end{aligned}$$

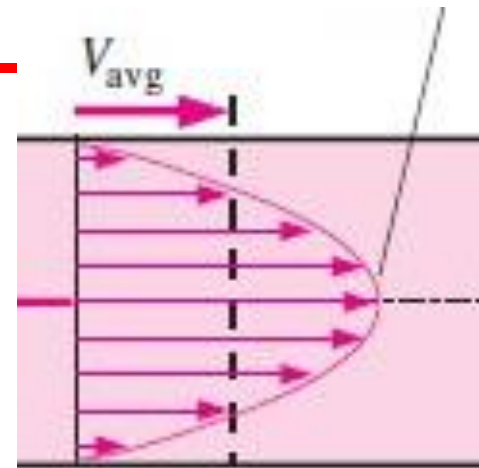
$$\text{Momentum of fluid /sec through section based on actual velocity} = \int_0^R 2\pi\rho u^2 r dr \quad 11$$

Viscous flow through circular pipes

Momentum correction factor

$$\beta = \frac{\text{Momentum per second based on actual velocity}}{\text{Momentum per second based on average velocity}}$$

$$u = \frac{1}{4\mu} \left(-\frac{\partial p}{\partial x} \right) (R^2 - r^2)$$



Momentum of fluid through ring /sec = mass $\times$ velocity

$$= \rho \times dQ \times u = \rho \times 2\pi r dr \times u \times u = 2\pi\rho u^2 r dr$$

$$\text{Momentum of fluid /sec through section based on actual velocity} = \int_0^R 2\pi\rho u^2 r dr$$

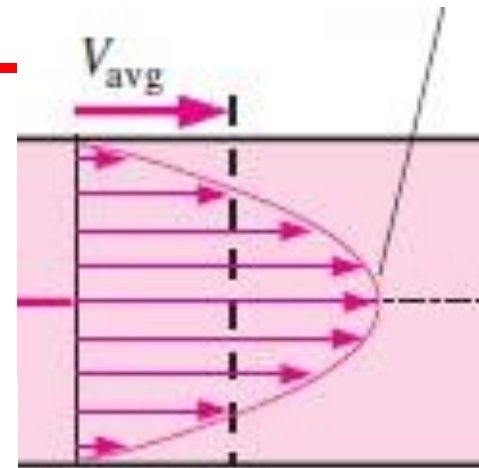
$$= \frac{\pi\rho}{48\mu^2} \left(\frac{\partial p}{\partial x} \right)^2 R^6$$

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Viscous flow through circular pipes

Momentum correction factor

$$\beta = \frac{\text{Momentum per second based on actual velocity}}{\text{Momentum per second based on average velocity}}$$



Momentum of the fluid per second based on average velocity

$$= \frac{\text{mass of fluid}}{\text{sec}} \times \text{average velocity}$$

$$= \rho A \bar{u} \times \bar{u} = \rho A \bar{u}^2$$

$$\bar{u} = \text{average velocity} = \frac{U_{\max}}{2}$$

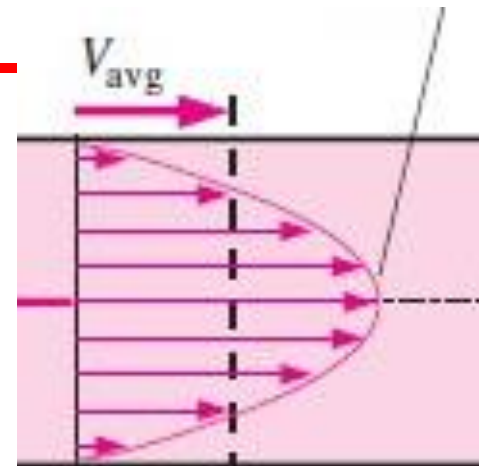
$$U_{\max} = \frac{1}{4\mu} \left(-\frac{\partial p}{\partial x} \right) R^2$$

$$\text{Momentum of fluid /sec through section based on average velocity} = \frac{\rho \pi \left(-\frac{\partial p}{\partial x} \right)^2 R^6}{64\mu^2}$$

Viscous flow through circular pipes

Momentum correction factor

$$\beta = \frac{\text{Momentum per second based on actual velocity}}{\text{Momentum per second based on average velocity}}$$



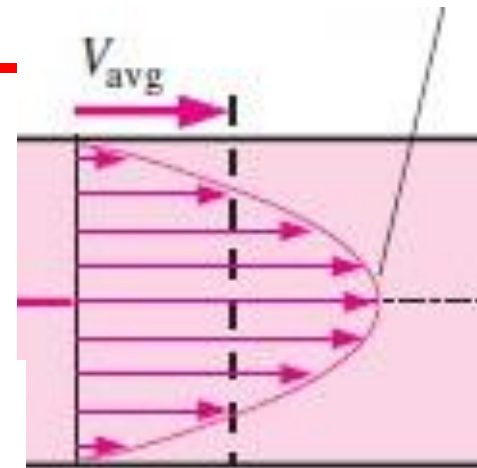
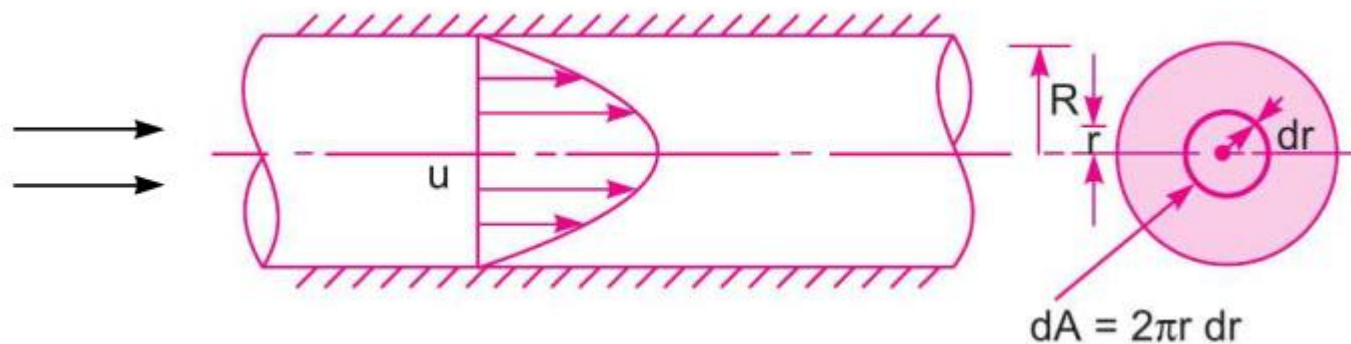
$$\begin{aligned} \text{Momentum of fluid /sec through section based on actual velocity} &= \frac{\pi \rho}{48 \mu^2} \left(\frac{\partial p}{\partial x} \right)^2 R^6 \\ \text{Momentum of fluid /sec through section based on average velocity} &= \frac{\rho \pi \left(-\frac{\partial p}{\partial x} \right)^2 R^6}{64 \mu^2} \end{aligned}$$

$$\beta = 4/3$$

Viscous flow through circular pipes

Kinetic energy correction factor

$$\alpha = \frac{\text{K.E./sec based on actual velocity}}{\text{K.E./sec based on average velocity}}$$



$$u = \frac{1}{4\mu} \left(-\frac{\partial p}{\partial x} \right) (R^2 - r^2)$$

$$\text{KE of fluid through ring /sec} = \frac{1}{2} \times \text{mass} \times u^2 = \frac{1}{2} \times \rho \times (u \times 2\pi r dr) \times u^2$$

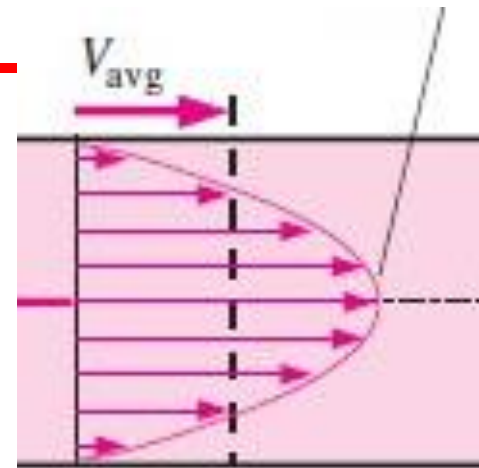
$$\text{KE of fluid /sec through section based on actual velocity} = \int_0^R \pi \rho r u^3 dr$$

Viscous flow through circular pipes

Kinetic energy correction factor

$$\alpha = \frac{\text{K.E./sec based on actual velocity}}{\text{K.E./sec based on average velocity}}$$

$$u = \frac{1}{4\mu} \left(-\frac{\partial p}{\partial x} \right) (R^2 - r^2)$$



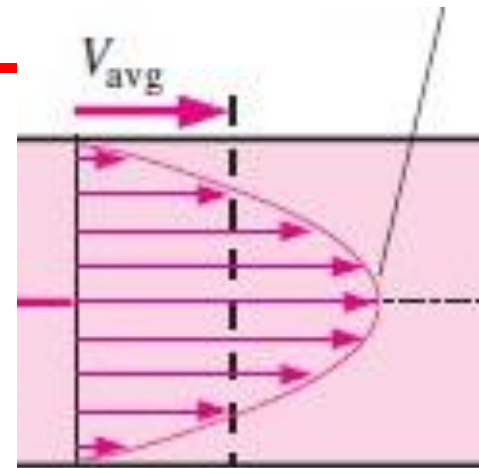
$$\text{KE of fluid through ring /sec} = \frac{1}{2} \times \text{mass} \times u^2 = \frac{1}{2} \times \rho \times (u \times 2\pi r dr) \times u^2$$

$$\begin{aligned} \text{KE of fluid /sec through section based on actual velocity} &= \int_0^R \pi \rho r u^3 dr \\ &= \frac{\pi \rho}{64\mu^3} \left(-\frac{\partial p}{\partial x} \right)^3 \frac{R^8}{8} \end{aligned}$$

Viscous flow through circular pipes

Kinetic energy correction factor

$$\alpha = \frac{\text{K.E./sec based on actual velocity}}{\text{K.E./sec based on average velocity}}$$



Kinetic energy of the flow based on average velocity

$$= \frac{1}{2} \times \text{mass} \times \bar{u}^2 = \frac{1}{2} \times \rho A \bar{u} \times \bar{u}^2 = \frac{1}{2} \times \rho A \bar{u}^3$$

$$\bar{u} = \text{average velocity} = \frac{U_{\max}}{2}$$

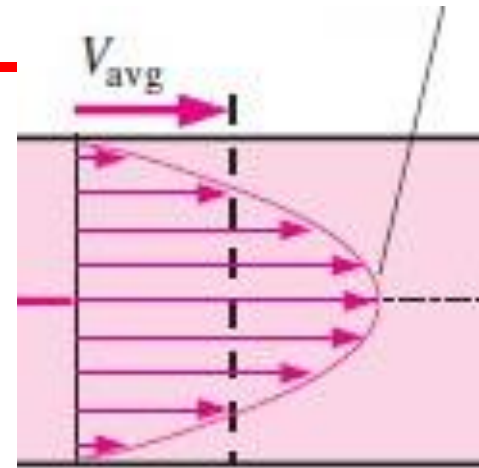
$$U_{\max} = \frac{1}{4\mu} \left(-\frac{\partial p}{\partial x} \right) R^2$$

$$\text{KE of fluid /sec through section based on average velocity} = \frac{\rho \pi}{128 \times 8\mu^3} \left(-\frac{\partial p}{\partial x} \right)^3 \times R^8$$

Viscous flow through circular pipes

Kinetic energy correction factor

$$\alpha = \frac{\text{K.E./sec based on actual velocity}}{\text{K.E./sec based on average velocity}}$$

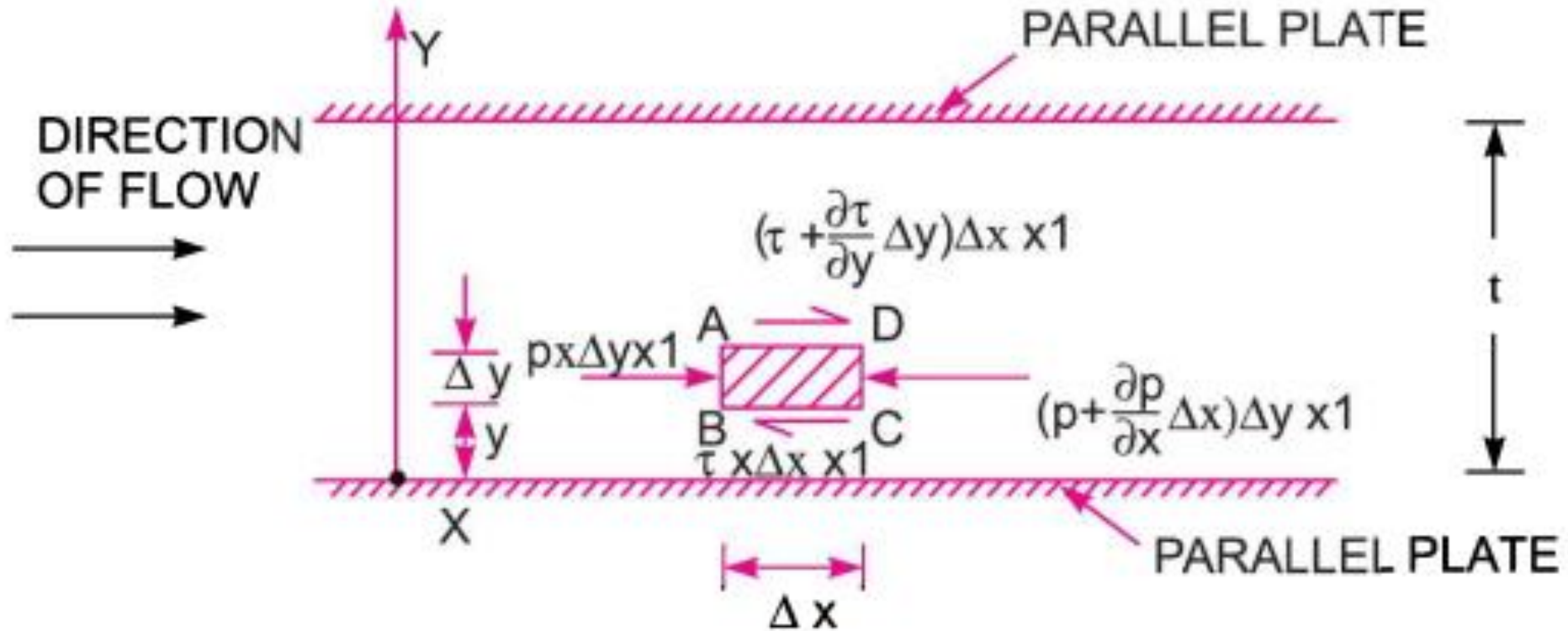


$$\text{KE of fluid /sec through section based on actual velocity} = \frac{\pi \rho}{64 \mu^3} \left(-\frac{\partial p}{\partial x} \right)^3 \frac{R^8}{8}$$

$$\text{KE of fluid /sec through section based on average velocity} = \frac{\rho \pi}{128 \times 8 \mu^3} \left(-\frac{\partial p}{\partial x} \right)^3 \times R^8$$

$$\alpha = 2$$

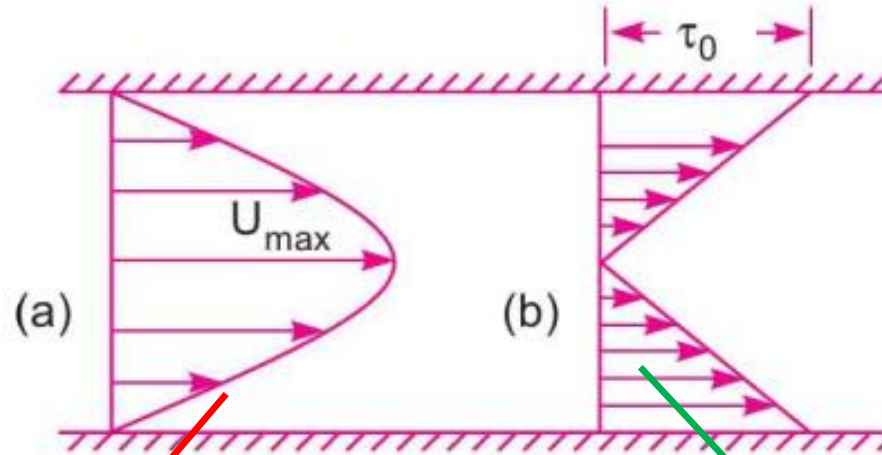
Viscous flow between two parallel plates



Viscous flow between two parallel plates



Shear stress and velocity distribution profiles



$$u = -\frac{1}{2\mu} \frac{\partial p}{\partial x} [ty - y^2]$$

$$\tau = -\frac{1}{2} \frac{\partial p}{\partial x} [t - 2y]$$

Viscous flow between two parallel plates



Relation between maximum velocity and mean velocity

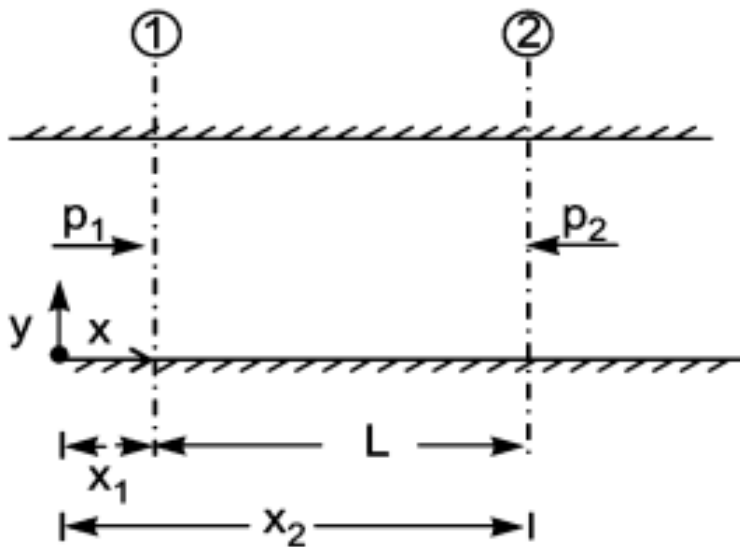
$$U_{\max} = -\frac{1}{2\mu} \frac{\partial p}{\partial x} \left[t \times \frac{t}{2} - \left(\frac{t}{2} \right)^2 \right]$$
$$= -\frac{1}{2\mu} \frac{\partial p}{\partial x} \left[\frac{t^2}{2} - \frac{t^2}{4} \right] = -\frac{1}{2\mu} \frac{\partial p}{\partial x} \frac{t^2}{4} = -\frac{1}{8\mu} \frac{\partial p}{\partial x} t^2$$
$$\bar{u} = \frac{Q}{\text{Area}} = -\frac{\frac{1}{12\mu} \frac{\partial p}{\partial x} t^3}{t \times 1} = -\frac{1}{12\mu} \frac{\partial p}{\partial x} t^2$$

$$\frac{U_{\max}}{\bar{u}} = \frac{-\frac{1}{8\mu} \frac{\partial p}{\partial x} t^2}{-\frac{1}{12\mu} \frac{\partial p}{\partial x} t^2} = \frac{12}{8} = \frac{3}{2}$$

Viscous flow between two parallel plates



Head loss estimation



$$h_f = \frac{p_1 - p_2}{\rho g} = \frac{12\mu\bar{u}L}{\rho g t^2}$$



THANK YOU